

A Novel Method for Identifying Pollen Allergens Using Chemically Modified Graphene

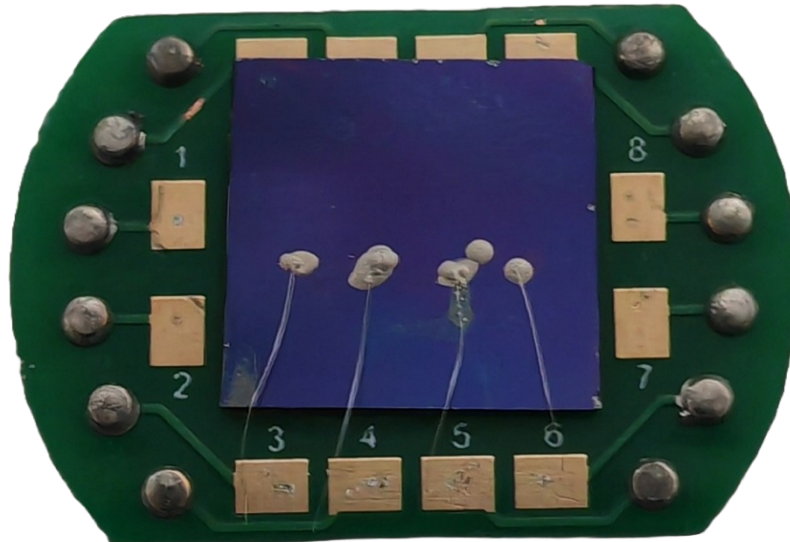


Fig.1 Graphene four-wire sensing electrodes, image taken by finalist, 2026

INTRODUCTION – POLLEN ALLERGY

In every **four** people in the world, **one** is allergic to pollen.

They experience symptoms from **allergic rhinitis to life-threatening anaphylactic shocks** through Thunderstorm Asthma.

In their daily routine, people with allergies rely heavily on **up-to-date** information about pollen counts (levels and concentrations) in their area, e.g., to adjust their medications.

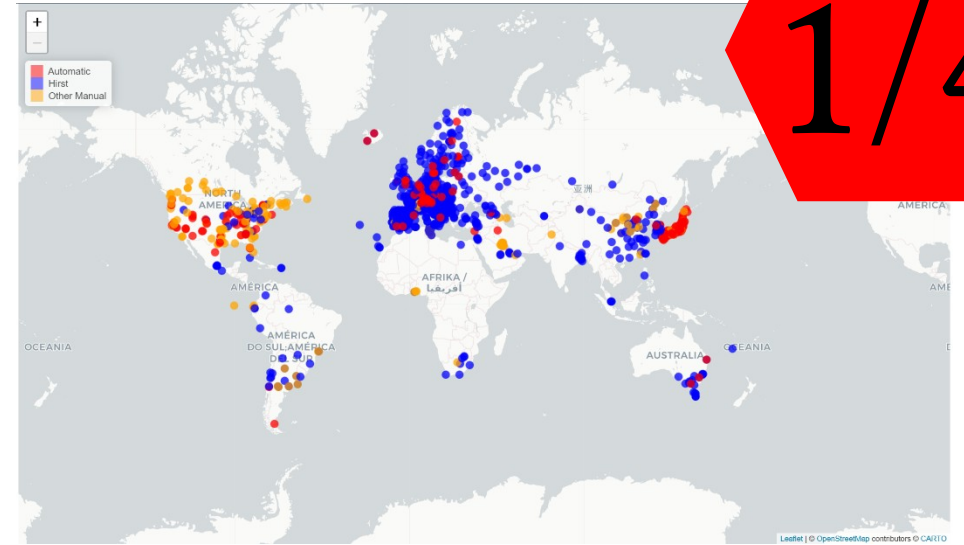


Fig.2: Worldwide Map of Pollen Monitoring Stations. Note: Data indicates a high density of Hirst/manual detection methods in China.

From Worldwide Map of Pollen Monitoring Stations, by Jose Oteros/ZAUM, 2026. Available at: https://oteros.shinyapps.io/pollen_map/



Fig.3: Thunderstorm Asthma. Note: Airborne pollen concentration surges due to moisture and electrical charges, leading to the rupture of pollen grains. Graphic created on April 10, 2026, using Google Gemini 3.



Fig.4: Finalist's Allergy Symptoms. Graphic created on April 10, 2026, using Google Gemini 3

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INTRODUCTION – CURRENT POLLEN DETECTION METHODS

However, traditional pollen detection methods are:

- **Too Slow** (count up to ~4 hrs, delay) → **daily pollen count relies on data from the previous day**
- **Not Precise** (detects pollen grains instead of proteins) → **vulnerable to Thunderstorm Asthma**
- **Too Costly** (**ML-based** pollen image recognition microscope costs ~\$ 23k)

Manual Pollen Sampling



Fig.5: Manual Pollen Sampling, by Beijing Shijitan Hospital, 2021. <https://zhuanlan.zhihu.com/p/355895368>

Finalist counting pollen grains under microscope

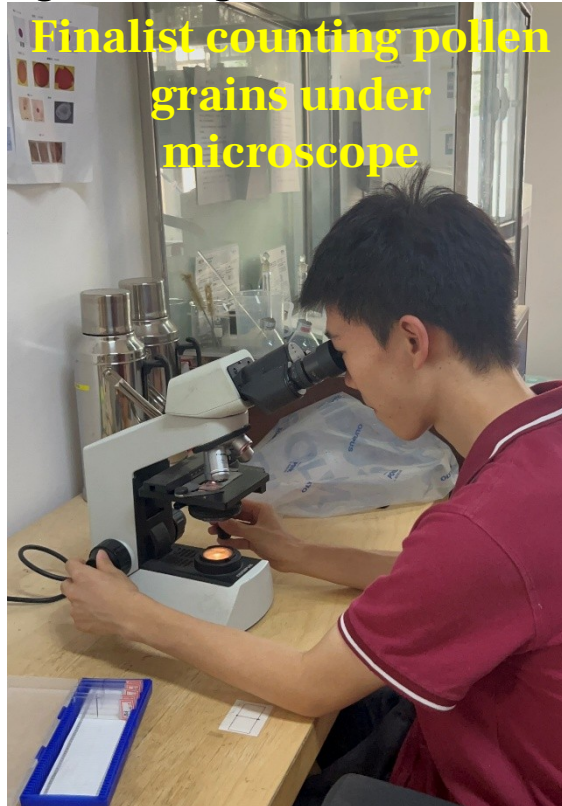


Fig.6: Counting and Recognizing Pollen Grains under Microscope, Photo taken by finalist, Beijing Shijitan Hospital, 2025.9

Skin Prick Test (SPT)

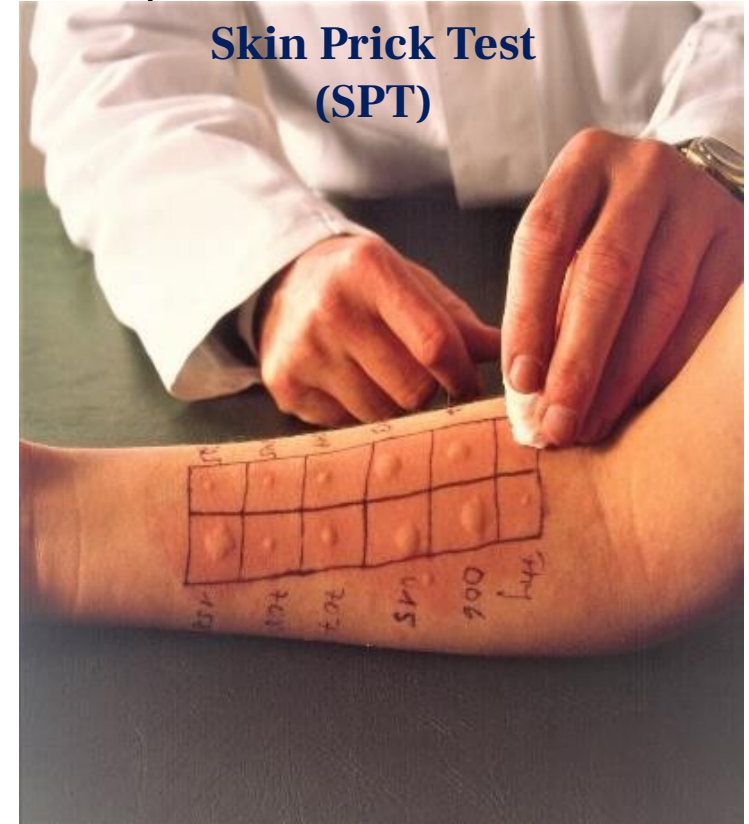


Fig.7: Positive reaction to a multi-allergen skin prick test. From *Allergy Test Allergietest*, by LOOK-foto/Image Professionals GmbH, 2005.

INTRODUCTION – GRAPHENE & ALLERGY

REACTION

Graphene

- 2D structure ♦ large surface area ♦ high carrier mobility
- **High electrical conductivity**, enabling ultra-sensitive detection of biomolecular binding and gas adsorption
- Detects **subtle surface interactions** (e.g., antigen binding, doping)
- **Doping mechanism**
- Electron withdrawal → **increases hole carrier density (p-type)**
- Leads to a **measurable increase in electrical**

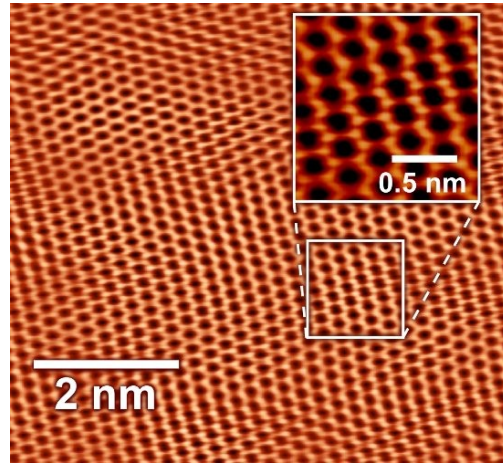


Figure X: Atomic-Resolution Scanning Tunneling Microscopy (STM) Image of Graphite. From Atomic-Resolution Images of Graphite, by Hannes C. Schniepp / Nano & Biomaterials Lab, 2009.

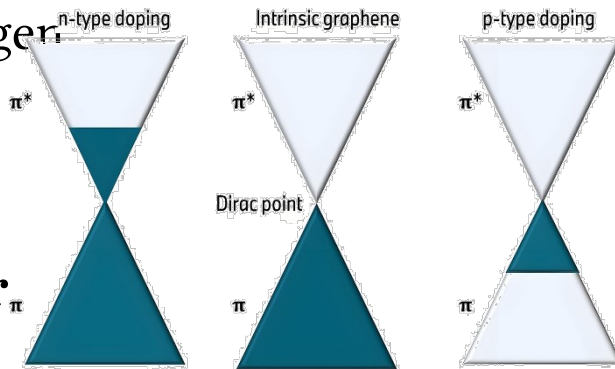
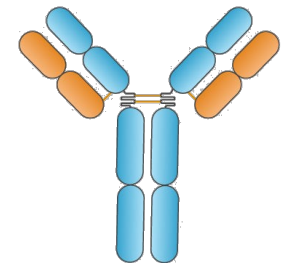


Figure X: Energy band structure and Dirac point of intrinsic and doped graphene. Graphic created by Finalist using Microsoft PowerPoint.

Human Allergy & IgG Recognition

- Allergic response:
 - a) Triggered when allergens (e.g., pollen proteins) are recognized by the immune system.
 - b) Primarily mediated by IgE antibodies, which leads to inflammatory reactions.
- IgG antibody
 - a) One of the five major immunoglobulins
 - b) Fab region binds specifically to antigen epitopes → **High specificity of antigen–antibody binding.**
 - c) This enables the sensor to selectively **recognize** and **differentiate** pollen proteins from impurities



IgG

Fig.8: Model illustrating the impact of IgG antibodies on Type I hypersensitivity. From On the role of antibody affinity in the IgE mediated allergic response, by Svatlana Starchenka, Martin Bachmann, et al., 2024.

Fig.9: Structure of the Immunoglobulin G (IgG) Antibody. From Antibody Isotypes IgG, IgA, IgM, IgE, IgD [Illustration], by Beta LifeScience, 2025.

METHODOLOGY – GRAPHENE CHEMICAL MODIFICATION

Modification & Sensing Interface

- **Sensing interface construction**
 - Graphene–PBASE–IgG functional structure for antigen capture
- **PBASE linker ($C_{24}H_{19}NO_4$):**
 - Pyrene group → π – π stacking with graphene
 - NHS ester group → covalent binding to IgG antibodies

Step	Interaction
Pyrene ↔ Graphene	π – π stacking
NHS ester ↔ IgG	Covalent amide bond

- **Antibody functionalization:**
 - **IgG antibodies** introduced in PBS solution
 - **Ragweed, birch IgG antibodies** immobilized on graphene

PBASE: 1-Pyrenebutyric Acid NHS Ester

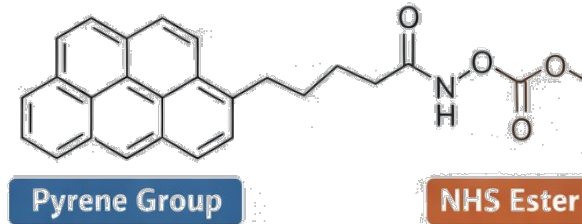


Figure 11: Chemical structure of 1-Pyrenebutyric acid NHS ester (PBASE). Graphic created on April 7, 2026, using ChatGPT-5 by OpenAI.

E.g.

- **Cypress** IgG – graphene
→ selectively detects **cypress** pollen proteins.
- **Birch** IgG – graphene
→ selectively detects **birch** pollen proteins.

Forms a selective sensing interface for pollen antigen binding

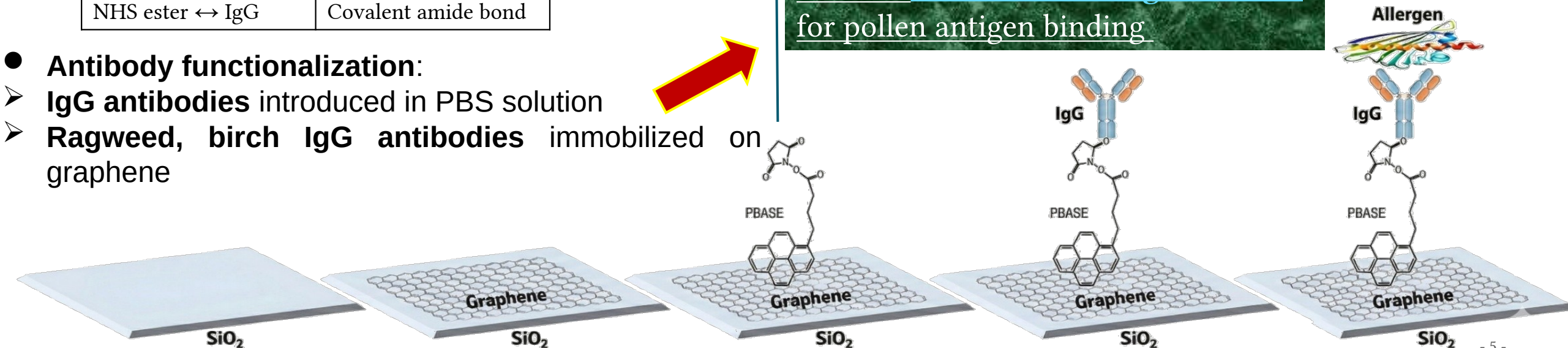


Fig.10: Graphene-FET biosensor functionalization and allergen detection process. Graphic created by Finalist using Microsoft PowerPoint.

METHODOLOGY – CHARACTERIZATION

Graphene Quality Verification

- Prepared via mechanical exfoliation & CVD

Raman

spectroscopy:

G' peak > G peak, indicating single-layer graphene

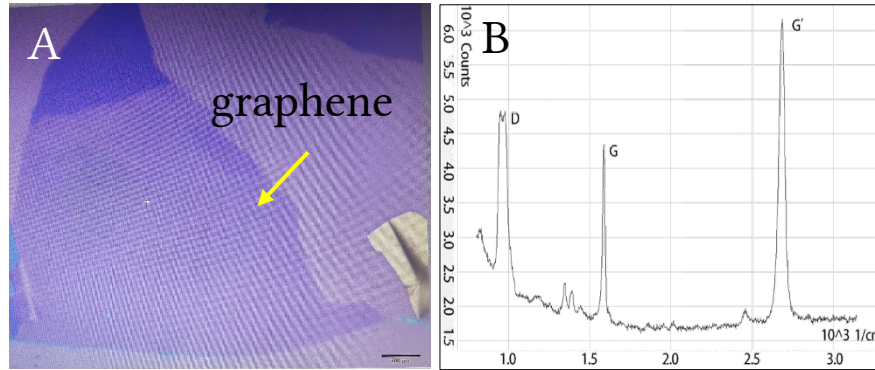


Fig.11: Characterization of mechanically exfoliated graphene on (A) Optical Microscopy image showing graphene flakes and (B) corresponding Raman spectrum with labeled characteristic G and 2D peaks. Images and data collected by Finalist using an optical microscope and Raman Spectrometer

PBASE Modification (AFM)

- Nanoscale protrusions (0.5–3 nm) observed on surface
- Correspond to PBASE molecules and aggregates
- Confirms successful surface functionalization

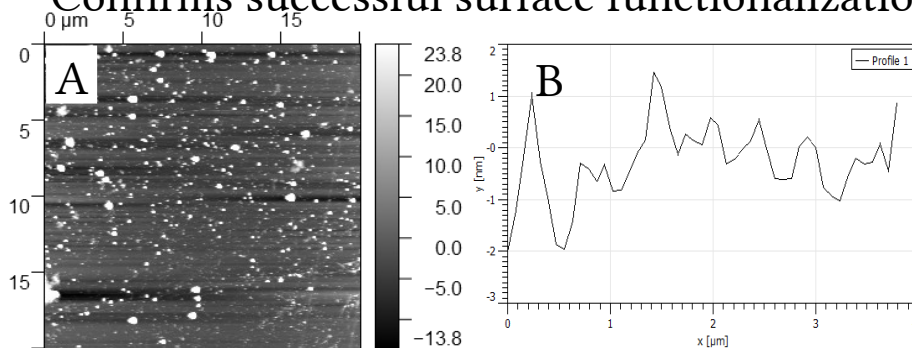
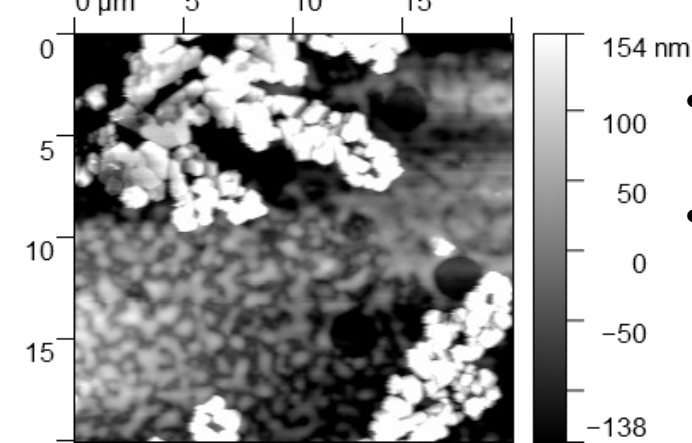


Fig.12: AFM height image of PBASE-modified graphene (A) and the corresponding line-scan height profile (B).

Images taken by the finalist using AFM

IgG Immobilization (AFM)



- Presence of uniform Y-shaped antibody structures
- Indicates stable and controlled antibody attachment

Fig.13 Pollen allergen antibody characterization, image taken by the finalist using AFM.

Pollen Protein Binding Verification

AFM shows spherical particles (8–12 nm diameter), which is consistent with pollen protein (e.g., Bet v 1) aggregates.

This confirms successful antigen binding on the sensor surface.

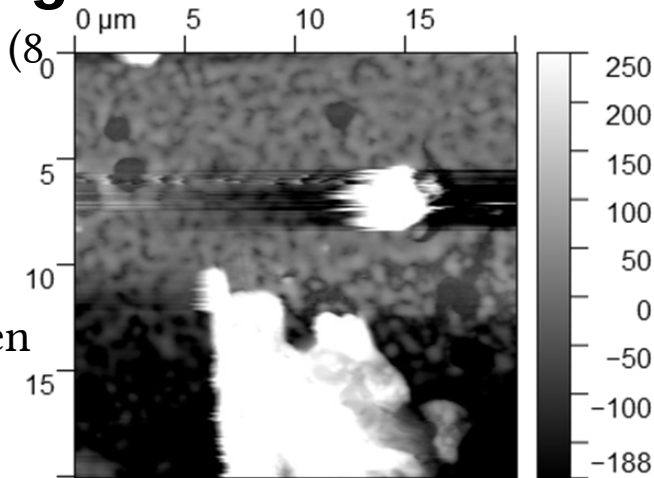


Fig.14 Bet V1 pollen protein antibody characterization, image taken by the finalist using AFM.

METHODOLOGY – RESISTANCE & ELECTROCHEMICAL

TESTING

● Resistance Measurement

- Four-probe method
 - Eliminates **contact resistance**
 - Enables **real-time**, low Limit of detection

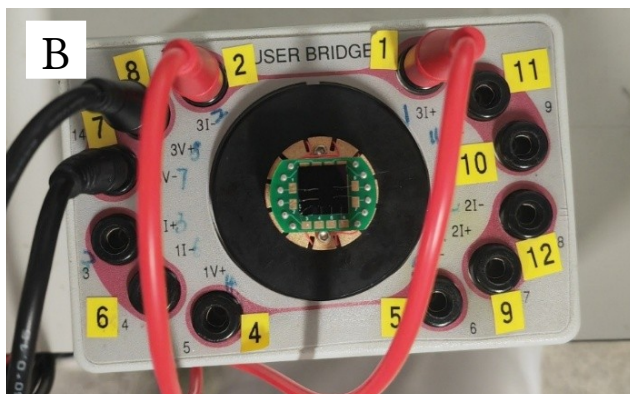
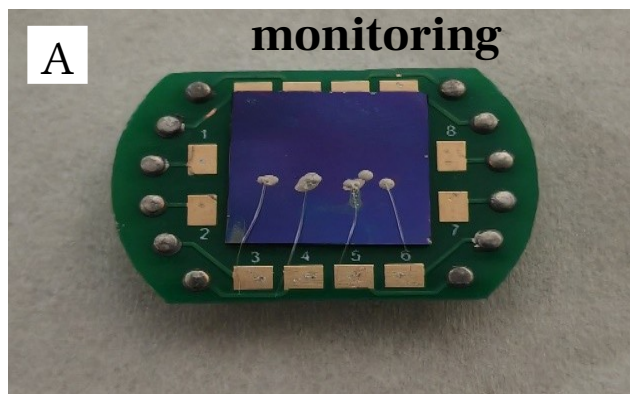


Fig.15: Graphene four-wire electrode design (A) and sensing base (B). Photo taken by Finalist.

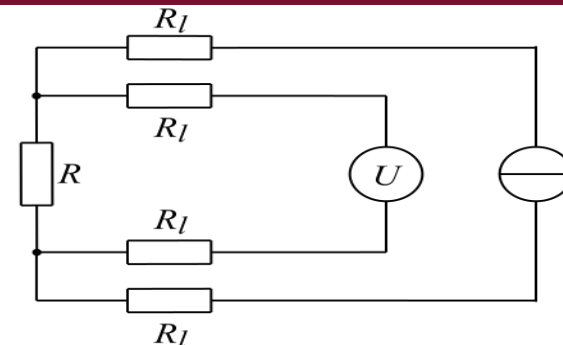


Fig.16: Four-wire (Kelvin) sensing circuit schematic. Diagram created by Finalist.

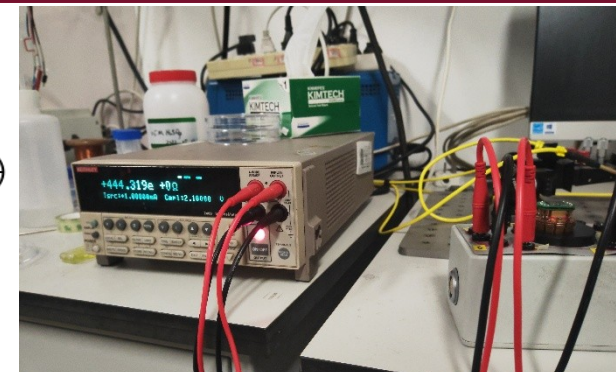


Fig.17: Digital source meter used for resistance sensing. Photograph taken by Finalist.

● Experimental Design

- **Group 1 (Selectivity Test)**
 - Antibody-functionalized graphene exposed to: **Specific** vs **non-specific** pollen proteins.
 - Measure **resistance change (ΔR)** before/after binding.
- **Group 2 (Quantification Test)**
 - Sensor exposed to **varying concentrations (1:5, 1:10, 1:15)** of target pollen proteins.
 - Measure **ΔR vs concentration** relationship.

● Electrochemical Validation

- **3 groups of Cyclic Voltammetry (CV) test:**
 - Confirm **binding behavior** and surface modification
 - Analyze **binding kinetics (stabilization time for engineering application)**



Fig.18: CHI760E Electrochemical Workstation for CV Validation. Photograph taken by Finalist.

RESULTS – RESISTANCE TESTING: TYPES AND CONCENTRATION

Group I

$$\rho = R \frac{A}{L}$$

where:

- ρ is the resistivity ($\Omega \cdot m$);
- R is the resistance (Ω);
- A is the cross-sectional area of the conductor (m^2);
- L is the length of the conductor (m).

Resistivity Normalization

Converting resistance to resistivity removes geometric variations (electrode length and cross-section) across graphene sensors.

Significance Test

At $\alpha = 0.05$, a one-tailed t-test shows post-doping resistivity change is significantly greater than pre-doping, confirming the doping effect.

Item	Variable 1	Variable 2
Mean	11.55092	4.007393
Variance	0.967988	0.404124
Observations	5	5
Pearson Correlation	0.92449	
Hypothesized Mean Difference	0	
Degrees of Freedom (df)	4	
t Stat	28.13471	
P(T ≤ t) one-tail	0.00063	
One-tail Critical t	2.919986	
P(T ≤ t) two-tail	0.001261	
Two-tail Critical t	4.302653	

Table 1: T-test (Paired Two-Sample for Means) evaluating sensor selectivity. This analysis verifies sensor’s ability to statistically distinguish between different pollen types. **Data analysis performed by Finalist using Microsoft Excel**

Group

Concentration Relationship

Trendline fitting reveals an exponential relationship between pollen concentration and resistivity change. Three fits ($R^2 \approx 1$) were combined to obtain:

$$y = 0.594e^{-239.3x}$$

1:5, 1:10, 1:15 dilution are used for AMB A1 protein

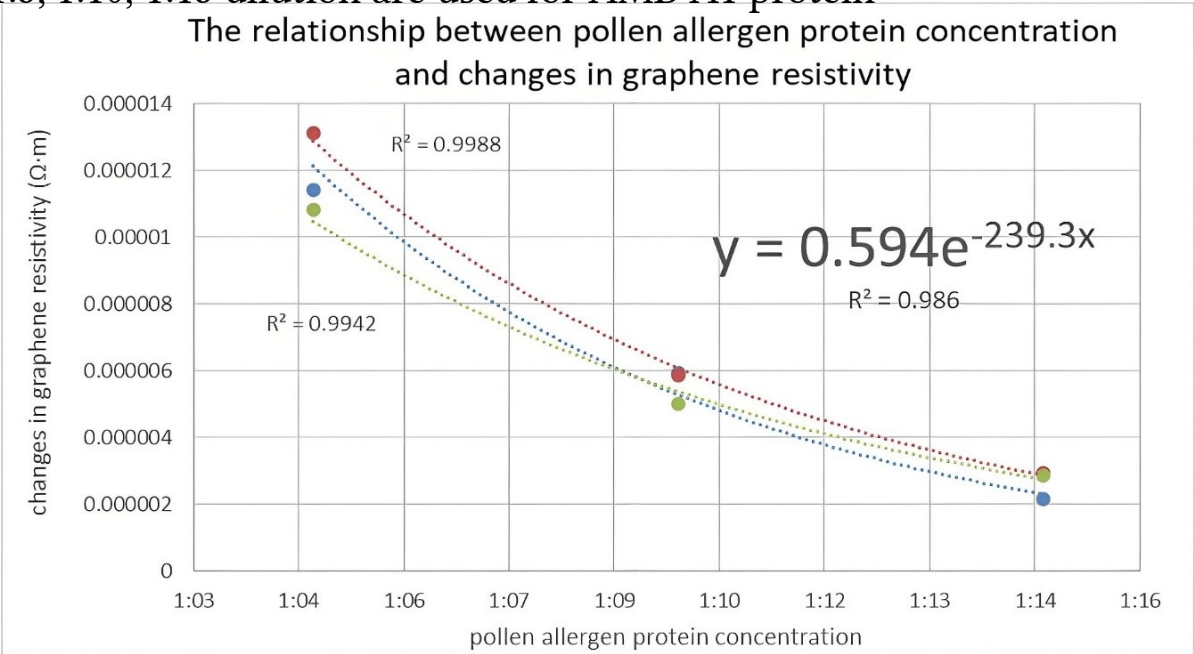


Fig.19: Correlation between pollen allergen protein concentration and graphene resistivity. The data demonstrates a significant exponential relationship, $R^2=0.98$. Mathematical modeling and data plotting performed by Finalist using Excel

RESULTS – ELECTROCHEMICAL (CV)

Specificity

- Low background signal
- PBS (blank): no redox peaks, $Q \approx 0$
- Selective detection
- Bet v 1 (target): strong redox peaks, $Q \sim 10^{-4} \text{ C}$
- Amb a 1 (non-specific): weaker peaks, $Q \sim 10^{-5} \text{ C}$

Fig. 20: CV curves for selectivity verification. Comparison of baseline PBS vs. varying dilutions (1:10 and 1:5) of Bet v1 allergen solution to verify sensor recognition. Data collected and analyzed by Finalist using CHI760E Workstation.

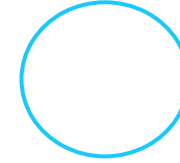
Quantitative Analysis

- Peak area (integrated charge) increases with **Bet v 1 concentration**
- Peak area calculated by:

$$Q = \int_{E_1}^{E_2} I(E) dE$$

Where **Q** = charge, **I(E)** = current at potential **E**

This establishes **concentration–signal relationship**



Binding Kinetics & Stability

Peak area increases rapidly in early cycles (~118% from cycle 1→3)

Stabilizes after ~7–9 cycles (~6 min)
(<13% change between later cycles)

→ **10 min chosen as standard sensing time**



Fig. 21: CV curves for pollen concentration quantification.

Sequential testing of PBS, 1:10, and 1:5 Bet v1. Data collected and analyzed by Finalist.

Fig. 23: Enlarged view of the positive potential region for sample S1_3B4_Bet v1 (1:10). Image and analysis generated by Finalist.

Fig. 22: Calibration curve of peak areas as a function of Bet v1 pollen concentration.

Statistical modeling and plotting performed by Finalist using Excel

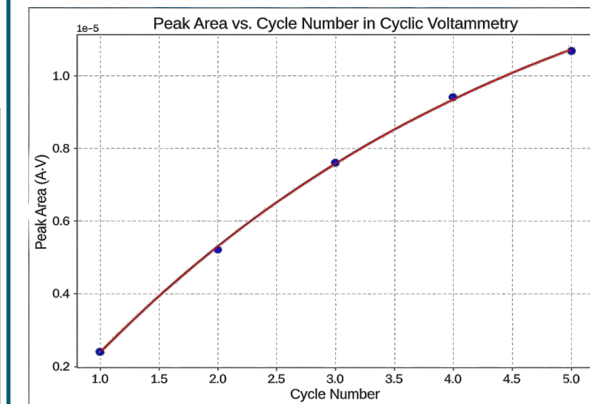
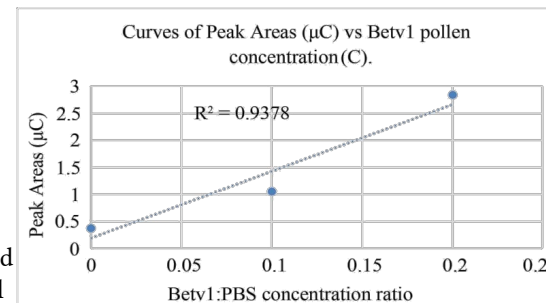


Fig. 24: Analysis of positive potential peak area stability over sequential cycles for sample S1_3B4. Data collected and analyzed by Finalist.

DISCUSSION

1. Sensitivity & Selectivity

- **High sensitivity:**
 - Detects $\Delta\rho$ down to $2.16 \times 10^{-6} \Omega \cdot m$ (1:15 dilution)
 - Estimated LOD: **38.6 $\mu g/mL$** (ragweed allergen)
 - Below typical immunization dose (150 μg protein)
- **Limitation & improvement:**
 - Variations in initial graphene resistivity leads to $\uparrow$ larger standard deviation, which limits LOD.
 - Higher-quality graphene could further lower LOD.
- **Strong selectivity:**
 - Specific vs non-specific response: **2.6 $\times$** higher signal $\rightarrow$ confirms **target-specific detection**

2. Exponential Relationship

- Exponential relationship ($R^2 = 0.98$)** between pollen concentration and resistivity $\rightarrow$ indicates signal is governed by **specific antigen-antibody binding** $\rightarrow$ consistent with **binding-site saturation behavior**
- Helps achieve **quantitative detection over extremely wide concentration range**

3. Electrochemical CV

- Reaction stabilizes at **~ 200 s**, which shows binding kinetics and optimal time window
- **10 min chosen as standard sensing time**
- However, electrochemical methods rely on bulky, high-cost instrumentation, making it not suitable for portable sensors.

4. Portability & Cost

- Traditional methods:**
- ML-based pollen microscopes $\approx$ **\$23,000**
 - Electrochemical workstations $>$ **\$10,000**
- This work(graphene biosensor): $<$ \$70 per device**

CONCLUSION & FUTURE WORKS

This study's chemically modified graphene biosensors achieve **selective, quantitative** detection of airborne allergen proteins.

Engineering pollen station

- **Portable system integration:** The graphene biosensor was combined with environmental sensors & ESP32 & AD5940 chip, achieving rapid ~10-minute testing workflow. It was also integrated with other environmental sensors(e.g., humidity sensor) to calculate the overall allergy risk.

Human allergy test

- **Broad diagnostic potential:** The same sensor could perform human allergy detection by sensing human IgG with pollen antigen doping first, replacing the current risky Skin Prick Test.



Figure 25: The 'Column of Trajan' Integrated Pollen Risk Sensor. Installed at school for real-time environmental monitoring. Photo taken by Finalist.

REFERENCE & ACKNOWLEDGMENTS

Reference

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Acknowledgments

Deepest thanks to **Professor Lin's research team for their generous guidance, and Beijing No.80 High School** for helping me to carry out these interdisciplinary studies. Our school-based station now serves 1,800 students every day, significantly reducing allergy-related absences by 60%.

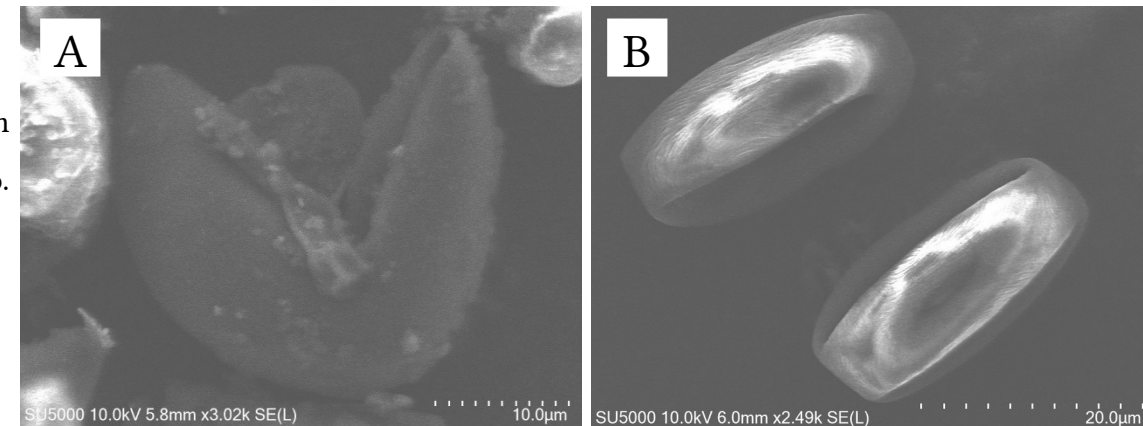


Fig. X: Scanning Electron Microscopy (SEM) analysis of pollen morphology. (A) Intact *Sorbaria sorbifolia* pollen grain and (B) ruptured Ragweed (*Ambrosia*) pollen grains showing internal cytoplasmic debris.
Images captured by Finalist using SEM.

A Novel Method for Identifying Pollen Allergens Using Chemically Modified Graphene

Naxi Tian, Beijing No.80 High School, Beijing, China

ENBM06

I. Problem

In every **four** people, **one** is allergic to pollen. They experience symptoms from **allergic rhinitis to life-threatening anaphylactic shocks**, and heavily depend on up-to-date pollen counts in their area. However, traditional pollen detection methods are:

- **Too Slow** (count up to ~4 hrs) → **daily pollen count rely on data from the previous day**
- **Not Precise** (detects pollen grains instead of proteins) → **vulnerable to Thunderstorm Asthma**
- **Too Costly** (ML-based pollen image recognition microscope **cost ~\$ 23k**)

Vision: Enable accurate, real-time airborne pollen monitoring



Fig 1 Pollen Trap,
Photo taken by Finalist at
Beijing Shijitan Hospital

II. Methodology

I. Graphene Functionalization & Characterization

- Modified with PBASE linker + IgG antibodies + pollen proteins(Bet v 1)
- Relies on specific **1 to 1** antibody-antigen binding
- Characterization verified: **AFM, SEM, Raman Spectroscopy**

II. Resistance sensing (Fig.2)

- Leverage graphene's **high carrier mobility** to transduce pollen binding into electrical signals

- **Four-probe sensing** to eliminate contact resistance

III. Electrochemical validation

- Cyclic Voltammetry (CV) test

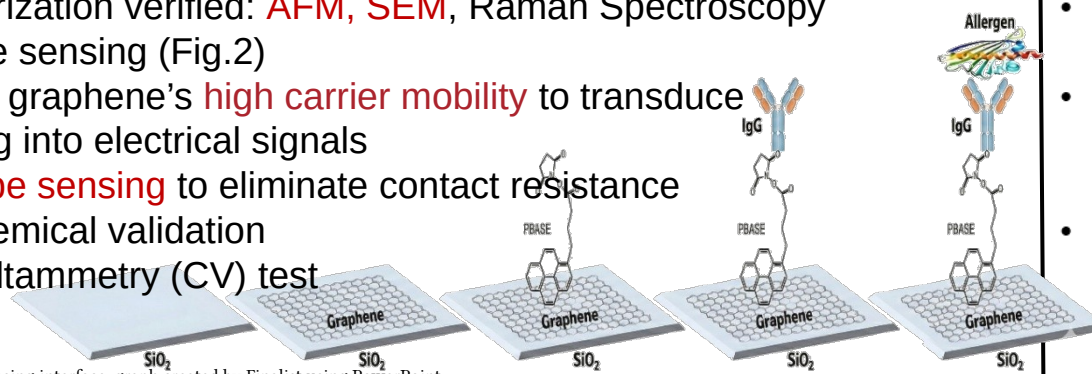


Fig.3 Graphene-PBASE-IgG sensing interface, graph created by Finalist using PowerPoint

III. Results

Selective detection ✓

- Graphene biosensor **significantly** differentiates between pollen proteins (one-tailed t-test $p < 0.05$)

Quantitative sensing relationship ✓

- Derived exponential relationship between Bet v1 pollen concentration and graphene resistivity change:
 $y = 0.549e^{-239.3x}$, ($R^2 > 0.98$)

Rapid stabilization from CV tests ✓

- CV shows pollen binding reaction stabilize after ~200 seconds

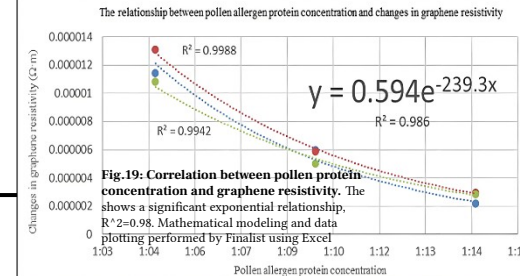


Fig.19: Correlation between pollen protein concentration and graphene resistivity. The shows a significant exponential relationship, $R^2 > 0.98$. Mathematical modeling and data plotting performed by Finalist using Excel

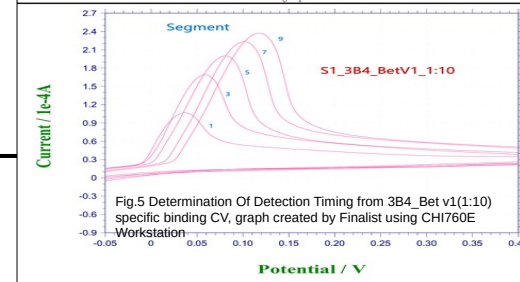


Fig.5 Determination Of Detection Timing from 3B4_Bet v1(1:10) specific binding CV, graph created by Finalist using CHI760E Workstation

IV. Conclusion

This study's chemically modified graphene biosensors achieve **selective, quantitative** detection of airborne allergen proteins.

- **High sensitivity:** Detects resistivity changes down to $2.16 \times 10^{-6} \Omega \cdot m$ (1:15 dilution).
- **Portable system integration:** combine environmental sensors & ESP32 & AD5940 chip, enabling rapid ~10-minute testing workflow.
- **Broad diagnostic potential:** Sensor could perform human allergy detection with antigen doping, suggesting it's value in allergy diagnosis and microorganisms detection.



Fig.7 "Column of Trajan"
Pollen Station at school, photo taken by Finalist